

MDA-EFSM Events:

Open()
Login()
IncorrectLogin()
IncorrectPin(int max)
CorrectPinBelowMin()
CorrectPinAboveMin()
Deposit()
BelowMinBalance()
AboveMinBalance()
Logout()
Balance()
Withdraw()
WithdrawBelowMinBalance()
NoFunds()
Lock()
IncorrectLock()
Unlock()
IncorrectUnlock()
Suspend()
Activate()
Close()

MDA-EFSM Actions:

A1: StoreData() // stores data from temporary area in data store
A2: IncorrectIdMsg() // displays incorrect ID message
A3: IncorrectPinMsg() // displays incorrect pin message
A4: TooManyAttemptsMsg() // display too many attempts message
A5: DisplayMenu() // display a menu with a list of transactions
A6: MakeDeposit() // makes deposit (increases *balance* by a value stored in *temp_d* in data store)
A7: DisplayBalance() // displays the current value of the *balance*
A8: PromptForPin() // prompts to enter pin
A9: MakeWithdraw() // makes withdraw (decreases *balance* by a value stored in *temp_w* data store)
A10: Penalty() // applies penalty (decreases *balance* by the amount of penalty)
A11: IncorrectLock Msg() // displays incorrect lock msg
A12: IncorrectUnlock Msg() // displays incorrect unlock msg
A13: NoFundsMsg() // Displays no sufficient funds msg

Operations of the Input Processor (ACCOUNT-1)

```

open (int p, int y, int a) {
// store p, y and a in temp data store
    ds->temp_p=p;
    ds->temp_y=y;
    ds->temp_a=a;
    m->Open();
}

pin (int x) {
    if (x==ds->pin) {
        if (ds->balance > 100)
            m->CorrectPinAboveMin ();
        else m->CorrectPinBelowMin();
    }
    else m->IncorrectPin(1)
}

deposit (int d) {
    ds->temp_d=d;
    m->Deposit();
    if (ds->balance>100)
        m->AboveMinBalance();
    else m->BelowMinBalance();
}

withdraw (int w) {
    ds->temp_w=w;
    m->withdraw();
    if ((ds->balance>100)
        m->AboveMinBalance();
    else m->WithdrawBelowMinBalance());
}

```

```

balance() {m->Balance();}

login (int y) {
    if (y==ds->uid)
        m->Login();
    else m->IncorrectLogin();
}

logout() {m->Logout();}

lock (int x) {
    if (ds->pin==x) m->Lock();
    else m->IncorrectLock();
}

unlock (int x) {
    if (x==ds->pin) {
        m->Unlock();
        if (ds->balance > 100)
            m->AboveMinBalance ();
        else m->BelowMinBalance();
    }
    else m->IncorrectUnlock();
}

```

Notice:

- m*: is a pointer to the MDA-EFSM object
- ds*: is a pointer to the Data Store DS-1 object

In the data store:

- balance*: contains the current balance
- pin*: contains the pin #
- id*: contains user id

Operations of the Input Processor (ACCOUNT-2)

```
OPEN (int p, int y, float a) {  
    // store p, y and a in temp data store  
    ds->temp_p=p;  
    ds->temp_y=y;  
    ds->temp_a=a;  
    m->Open();  
}
```

```
PIN (int x) {  
    if (x==ds->pin)  
        m->CorrectPinAboveMin ();  
    else m->IncorrectPin(2)  
}
```

```
DEPOSIT (float d) {  
    ds->temp_d=d;  
    m->Deposit();  
}
```

```
WITHDRAW (float w) {  
    ds->temp_w=w;  
    if (ds->balance>0)  
        m->Withdraw();  
    m-> AboveMinBalance()  
    else m->NoFunds();  
}
```

```
BALANCE() {m->Balance();}
```

```
LOGIN (int y) {  
    if (y==ds->uid)  
        m->Login();  
    else m->IncorrectLogin();  
}
```

```
LOGOUT() {m->Logout();}
```

```
suspend () {  
    m->Suspend();  
}
```

```
activate () {  
    m->Activate();  
}
```

```
close () {  
    m->Close();  
}
```

Notice:

m: is a pointer to the MDA-EFSM object

ds: is a pointer to the Data Store DS-2 object

In the data store:

balance: contains the current balance

pin: contains the pin #

id: contains user id